LAB 7 – Implementation of UI Components & User Controls

AIM: To design and implement essential UI components such as forms, buttons, lists, cards, navigation bars, alerts, and interactive controls to enhance user experience and improve app usability.

Practical Description (Detailed Format)

1. Introduction

User Interface (UI) plays a critical role in making apps effective, user-friendly, and visually appealing. In this lab, students will:

- Build functional forms
- Display items using lists/cards
- Add interactive components (buttons, switches, dropdowns)
- Implement dialogs, alerts, snackbars/toasts
- Apply basic theming and styling
- Use navigation bars or drawers

This practical ensures the application becomes **visually structured and interactive**, preparing it for navigation (Lab 8), API features (Lab 9), and advanced modules (Lab 11).

2. Practical Objectives

By the end of this practical, students will be able to:

- Use standard UI components in mobile apps
- Build clean and responsive screen layouts
- Display dynamic data using lists and cards
- Implement navigation components (drawer, bottom nav)
- Use alerts, dialogs, and snackbars for user feedback
- Enhance app usability with proper spacing, alignment, and styling

3. What to Do / How to Do

STEP 1: Identify Screens That Need UI Elements

Examples:

- Login Screen (forms, buttons)
- Registration Screen (multiple inputs)
- Dashboard (cards, buttons)
- CRUD list screen (list tiles, cards)
- Add/Edit Screens (text inputs, dropdowns)
- Profile Page (image, icons, text blocks)

List each screen and the UI components needed.

STEP 2: Build Forms Using Input Components

Components may include:

- Text fields
- Dropdown menus
- Checkboxes
- Radio buttons
- Date pickers
- Toggle switches
- Password visibility toggle

Guidelines:

✓ Provide labels and hints

✓ Validate all critical fields

✓ Display error text when incorrect input is entered

STEP 3: Add Buttons & Action Components

Use:

- Elevated / Filled buttons
- Floating Action Button (FAB)
- Icon buttons for edit/delete
- Back navigation icon in AppBar

Best Practices:

✓ Follow primary color scheme

✓ Use icons to enhance clarity

✓ Maintain consistent padding

STEP 4: Implement List & Card Views

Used for displaying dynamic data (CRUD output).

For lists:

- ListView / RecyclerView
- LazyColumn (Jetpack Compose)
- ListView.builder (Flutter)

For cards:

- Card widget / MaterialCardView
- Contain Title, Subtitle, Optional Image, Icons

Students must create at least **one styled list** and **one card layout**.

STEP 5: Implement Navigation UI Elements

Student must implement at least one:

Navigation Drawer

- Menu items like Home, Profile, Settings, Logout

Bottom Navigation Bar

- Tabs such as Dashboard, Items, Notifications

Top AppBar

- Title
- Action icons

STEP 6: Add Alerts, Dialogs & Snackbars

Use for:

- Delete confirmation
- Success messages
- Error messages

Examples:

- AlertDialog (Android)
- showDialog() (Flutter)
- Snackbar for success feedback

STEP 7: Apply Basic UI Styling

Students must refine:

- Colors
- Typography
- Padding & spacing
- Alignment
- Shape (rounded corners)

No need for final design polish, but UI must be **clean and readable**.

4. Expected Outcome

After completing this practical, students will have:

- A clean, structured, and interactive UI
- Functioning forms and lists
- Alerts, dialogs, and navigation controls
- Improved usability and design consistency
- Ready UI structure for integration with CRUD & API in next labs

5. Submission Requirements

✓ 1. Screenshots

Include:

- Forms
- List view
- Cards

- Navigation bar or drawer
- Dialogs / snackbars

✓ 2. Code Snippets (NOT full code)

Provide:

- One form code
- One list/card code
- Navigation UI code

✓ 3. GitHub Commit Link

✓ 4. Small Report (PDF)

Filename:

`RollNo_Name_Lab7_UIComponents.pdf`

6. Evaluation Rubric (10 Marks)

Criteria	Marks
UI components implemented correctly	3
Quality of layout & alignment	3
Correct use of navigation UI	2
Screenshots & documentation	1
Timely submission	1
Total	**10 Marks**